COP 4020: Haskell Assignment Calculating Π

There are different formulas giving pi approximations.
Implement those formulas in Haskell. The expected output is given next page.

a) Short pi= $\frac{1}{4}\pi = 4\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right).$

b) Simple Pi = $$\frac{\pi}{4} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} = 1 - \frac{1}{3} + \frac{1}{5} - \ldots$$

c) Wells pi $$\frac{1}{8}\pi^2 = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots$$ =

d) Euler pi $$\pi = \sqrt{6\left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \ldots\right)},$$ =

e) Wallis formula = $$\frac{\pi}{2} = \prod_{n=1}^{\infty}\left[\frac{(2n)^2}{(2n-1)(2n+1)}\right] = \frac{2\cdot 2}{1\cdot 3}\,\frac{4\cdot 4}{3\cdot 5}\,\frac{6\cdot 6}{5\cdot 7}\cdots$$

Hint: Writing a function which calculates $1/n^2$ series might be useful.
You can use an online compiler to implement this assignment
(https://play.haskell.org/)

While calculating the functions (besides short pi) please iterate 200 terms. An example output is below. The exact numbers are not given in the example. After stating each method's name, please print the calculated pi for that method. Please answer this question in your submission as a comment:

Apart from *Short Pi*, which pi calculation method is giving the most precise output? Please specify this in your submission as **a comment** inside the main function.

Do not forget to add as many comments as possible. Please also add header comments to your file. Header comment should include your name, PID and what this assignment is about. Also add the online compiler link to the header comment. atan function is used for calculating arc tangent in Haskell. https://www.tutorialspoint.com/haskell-program-to-find-the-arctangent-of-the-given-value

Simple Pi

3.1....

Wells Pi

3.1....

short Pi

3.1.....

euler Pi

3.1....

wallisSum Pi

3.1....

Rubric:

If code compiles 10 pts

Header comment includes name/PID and link 5 pts

Meaningful comments in the code 5 pts

Answer of the question: 10 pts (if even one output is missing, no points given for this part)

Simple Pi calculation 15 pts

If no output but some implementation 5 pts

Wells Pi calculation output 15 pts

If no output but some implementation 5 pts

short Pi calculation output 10 pts

If no output but some implementation 5 pts

euler Pi calculation output 15 pts

If no output but some implementation 5 pts

wallisSum Pi calculation output 15 pts

If no output but some implementation 5 pts